Name: _____

WORKSHEET

Vectors in polar form

1 A vector has magnitude 4 and direction 120° from the positive x-axis. Show five different representations of the vector.

2 Use a ruler and protractor to draw each of the following position vectors on the same set of axes.

a $(3, 45°)$

b $(5, 90°)$

c $(2, 135°)$

d $(5, 180°)$

e $(1, 225°)$

f $(4, 270°)$

g $(3.5, 315°)$

h $(4.5, 0°)$

3 Use a ruler and protractor to draw each of the following position vectors on the same set of axes.

a $(16, 30°)$

b $(20, 275°)$

c $(12, 160°)$

d $(8, 300°)$

e $(16, 87°)$

f $(4, 204°)$

g $(14.4, 345°)$

h $(17.2, 251°)$

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4 Draw the following vectors on the same set of axes. The vectors are given in polar form, with the angle expressed in radians.

a $\left(4, \frac{\pi}{4}\right)$

b $\left(5, \frac{\pi}{3}\right)$

c $\left(3, \frac{\pi}{2}\right)$

d $\left(4, \frac{5\pi}{4}\right)$

e $\left(5, \frac{7\pi}{4}\right)$

f $\left(2.5, \frac{3\pi}{2}\right)$

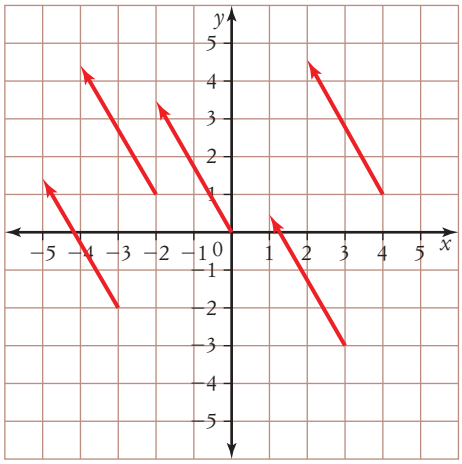
g $\left(4.5, \frac{2\pi}{3}\right)$

h $\left(3.5, \frac{17\pi}{12}\right)$

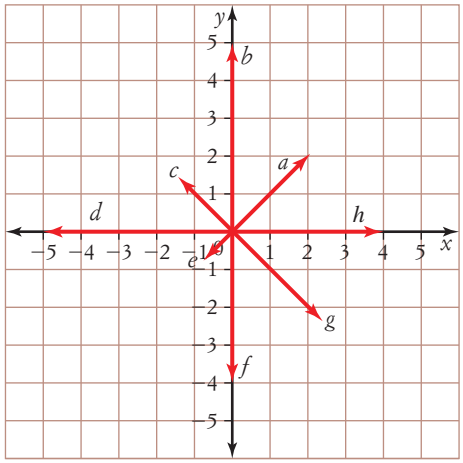
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Answers

1 1 unit = 1 cm

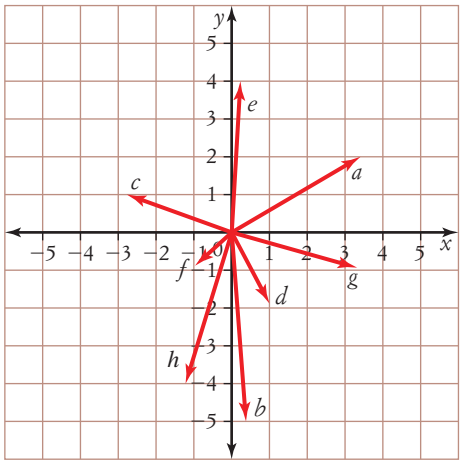


2 1 unit = 1 cm

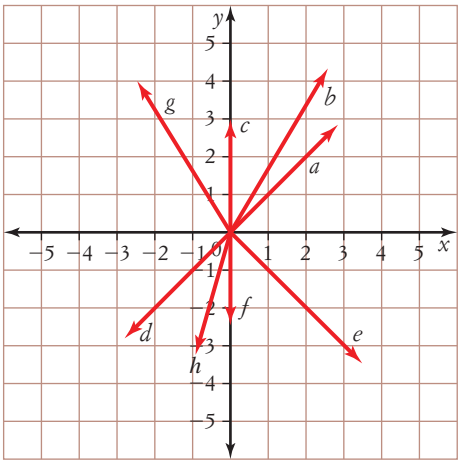


3 1 : 4

1 unit = 4 cm



4 1 unit = 1 cm



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